Polysaccharide

At the end of this topic, students will be able to;

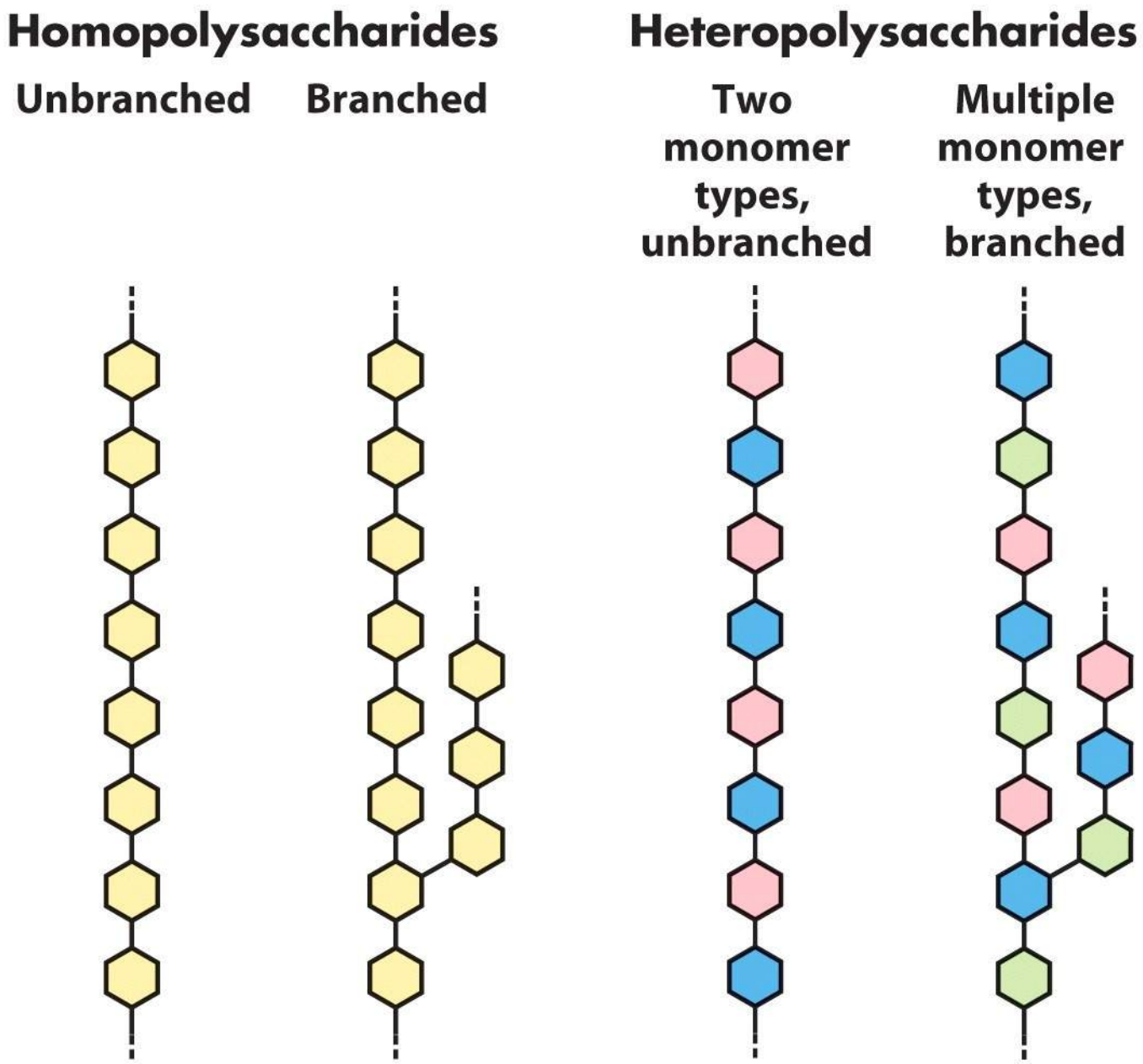
- Understand the basics of carbohydrates
- Define the polysaccharides
- Structure and classification of different polysaccharides with biochemistry.
- Functions and biomedical importance of polysaccharides

POLYSACCHARIDES

- Polymeric carbohydrate molecules composed of long chains of monosaccharide units bound together by glycosidic linkages.
- On hydrolysis give the constituent monosaccharides or oligosaccharides.
- ***Homopolysacchrides***: Starch, Dextrin, Dextrans, Inulin, Glycogen, Cellulose, Chitin
- ***Homopolysacchrides***: Hyaluronic acid, Mucopolysaccharides, Chondroitin sulphate, Heparin, Dermatan Sulphate, etc

Polysaccharides

- **Polysaccharides are long chains of many monosaccharides joined together by glycosidic bonds.**
- **There are three important polysaccharides:**
  - **Starch**
  - **Glycogen**
  - **Cellulose**

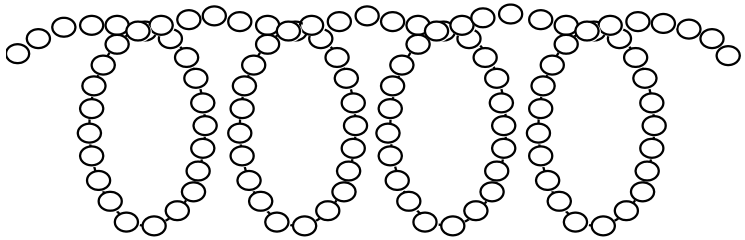

Homopolysaccharides
Unbranched
Branched
Heteropolysaccharides
Two monomer types, unbranched
Multiple monomer types, branched

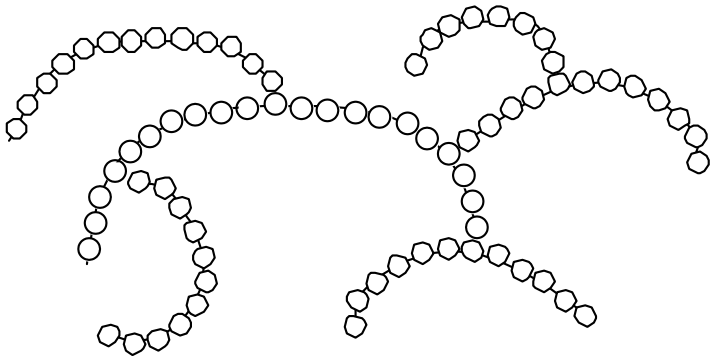
Starch

- This is the plant storage polysaccharide.
- It is insoluble and forms starch granules inside many plant cells.
- It is not a pure substance, but is a mixture of ***amylose*** and **amylopectin**

Amylose and amylopectin

- **Amylose is simply poly-(1-4) glucose, so is a straight chain. In fact the chain is floppy, and it tends to coil up into a helix.**



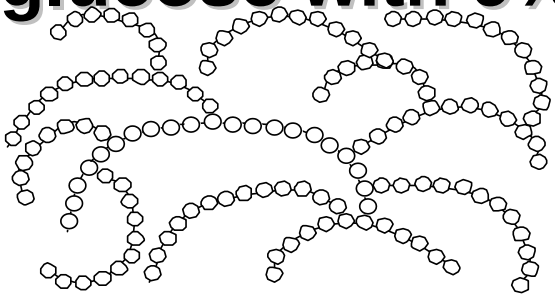


- **Amylopectin is poly(1-4) glucose with about 4% (1-6) branches. This gives it a more open molecular structure than amylose. Because it has more ends, it can be broken more quickly than amylose by amylase enzymes.**

- **Both amylose and amylopectin are broken down by the enzyme amylase into maltose, though at different rates.**
- **The enzyme can only break down the 1-4 bond NOT the 1-6 so starch is broken down into glucose and a disaccharide**

Glycogen

- This is similar in structure to amylopectin.
- It is poly (1-4) glucose with 9% (1-6) branches.



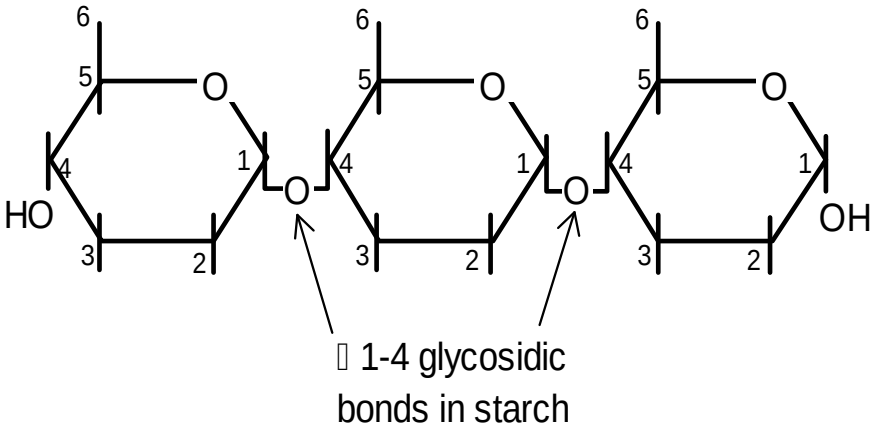
- It is made by animals as their storage polysaccharide, and is found mainly in muscle and liver.

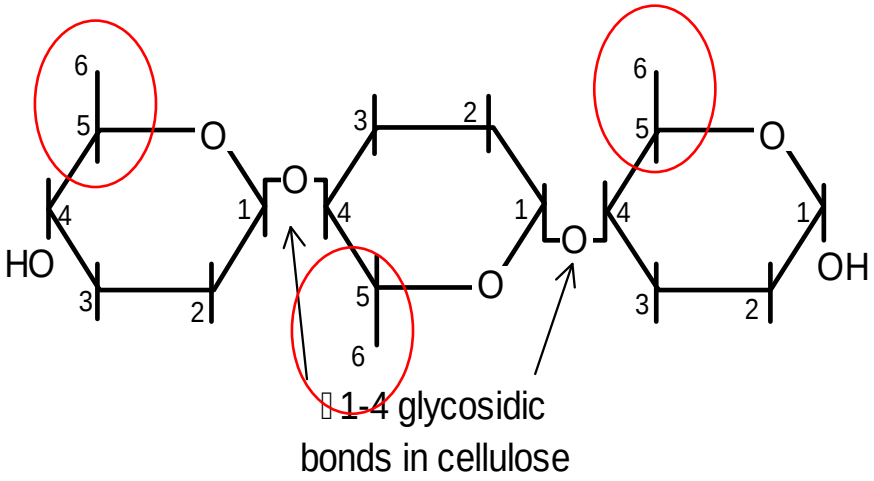
- **Because it is so highly branched, it can be mobilised (broken down to glucose for energy) very quickly.**

- **It is broken down to glucose by the enzyme glycogen phosphorylase.**

Cellulose

- **This is only found in plants, where it is the main component of cell walls.**
- **It is poly (1-4) glucose, but with a different isomer of glucose.**

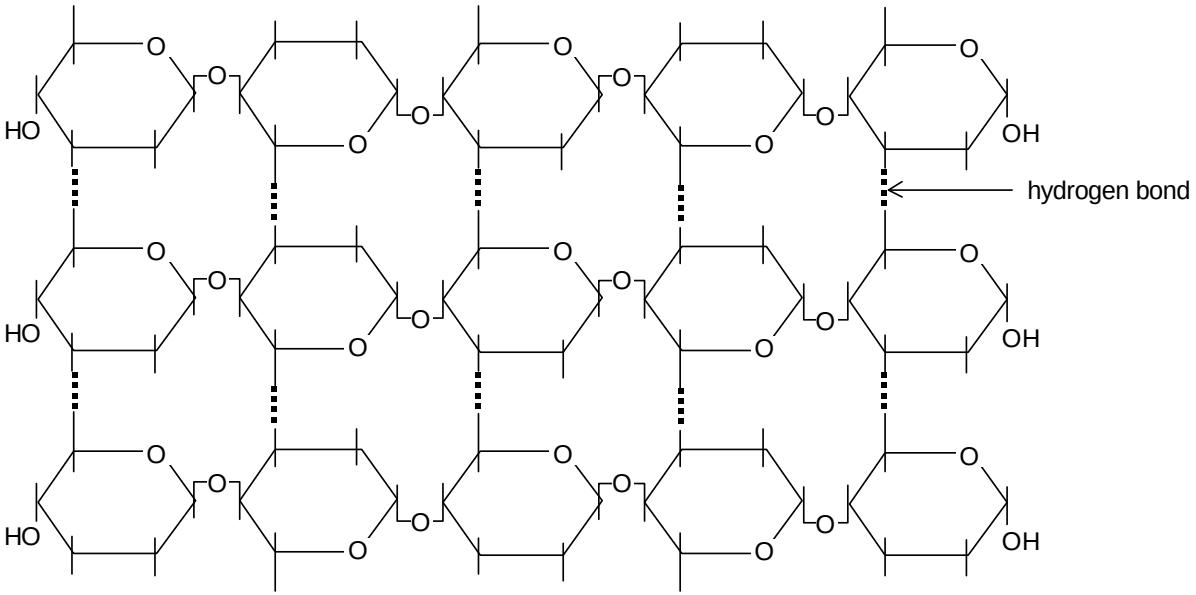




- **Starch and glycogen contain α-glucose, in which the hydroxyl group on carbon 1 sticks down from the ring, while cellulose contains β-glucose, in which the hydroxyl group on carbon 1 sticks up.**
- **This means that in a chain alternate glucose molecules are inverted.**

Starch and Cellulose

- **Whereas the α1-4 glucose polymer in starch coils up to form granules, the β1-4 glucose polymer in cellulose forms straight chains.**
- **Hundreds of these chains are linked together by hydrogen bonds to form cellulose microfibrils.**



- **These microfibrils are very strong and rigid, and give strength to plant cells, and therefore to young plants and also to materials such as paper, cotton and sellotape.**

Other polysaccharides

- **Chitin (poly glucose amine), found in fungal cell walls and the exoskeletons of insects.**
- **Cellose (poly 1-3 glucose), found in the walls of phloem tubes.**
- **Dextran (poly 1-2, 1-3 and 1-4 glucose), the storage polysaccharide in fungi and bacteria.**
- **Inulin (poly fructose), a plant food store.**

- **Pectin (poly galactose uronate), found in plant cell walls.**
- **Agar (poly galactose sulphate), found in algae and used to make agar plates.**
- **Murein (a sugar-peptide polymer), found in bacterial cell walls.**
- **Lignin (a complex polymer), found in the walls of xylem cells, is the main component of wood.**

Extracellular Matrix Proteolgycan

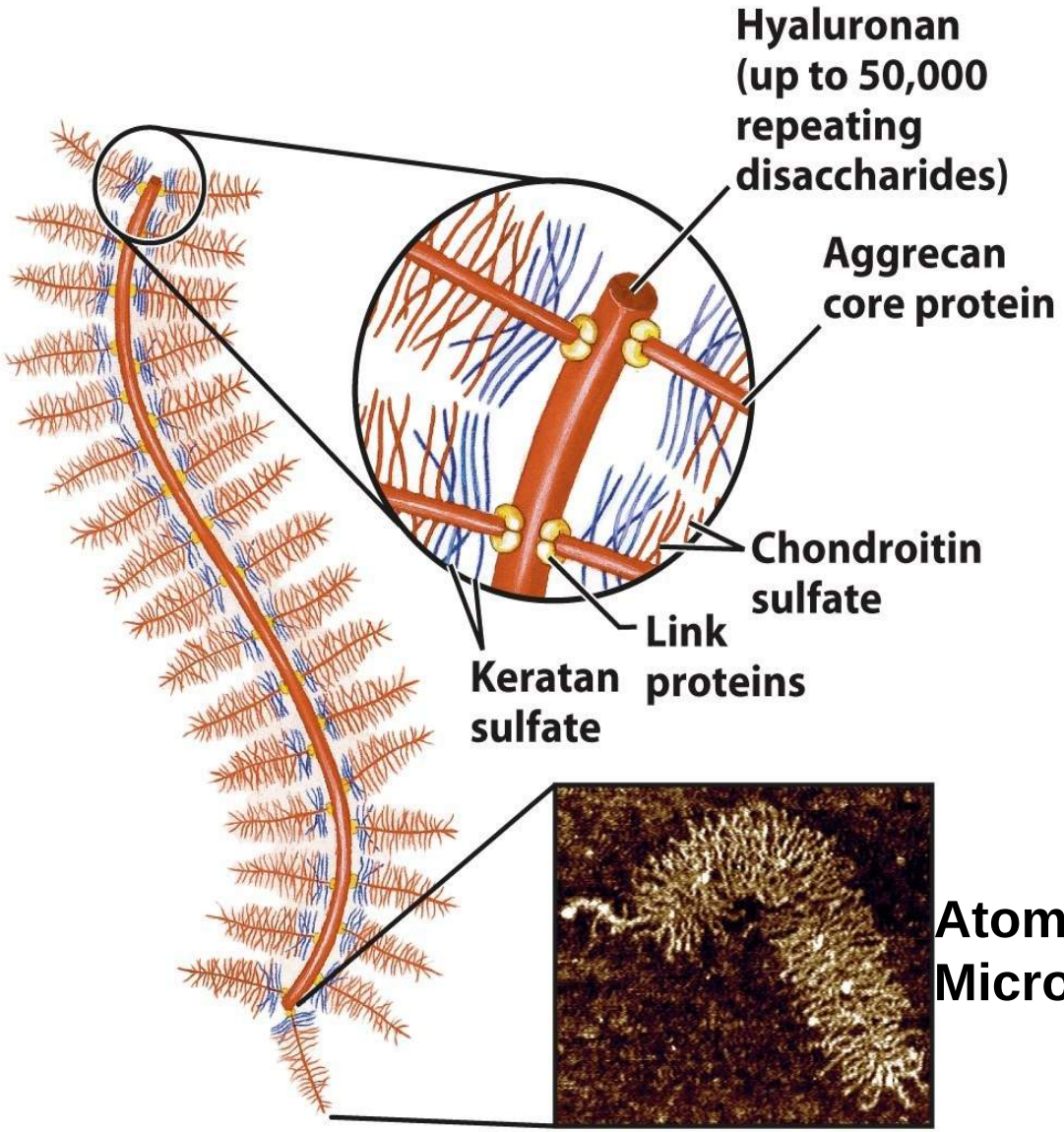
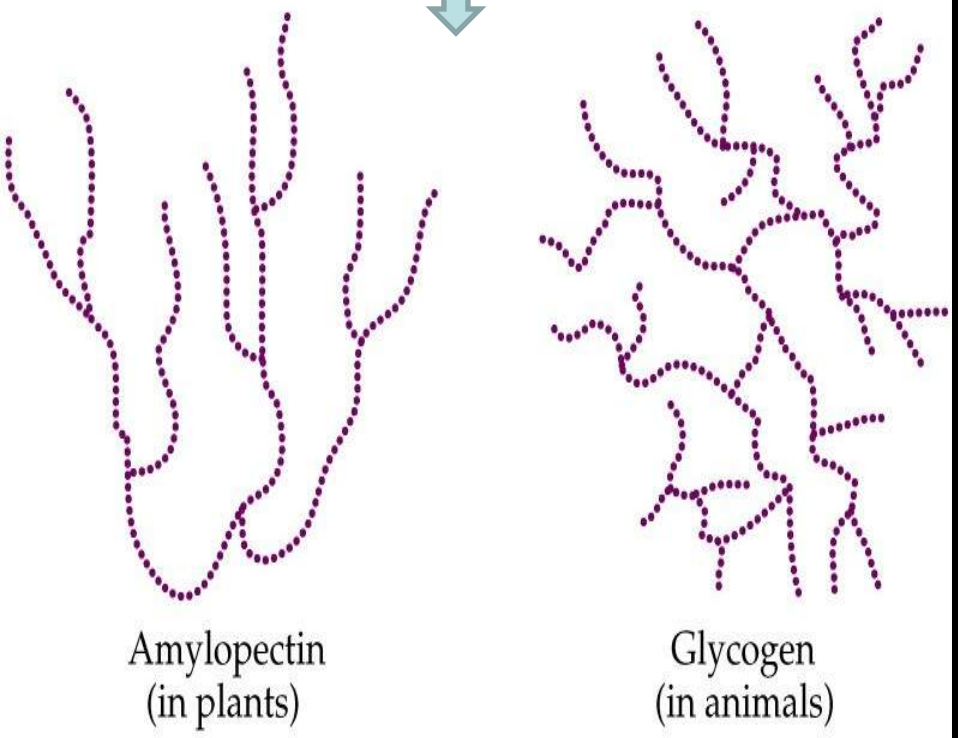
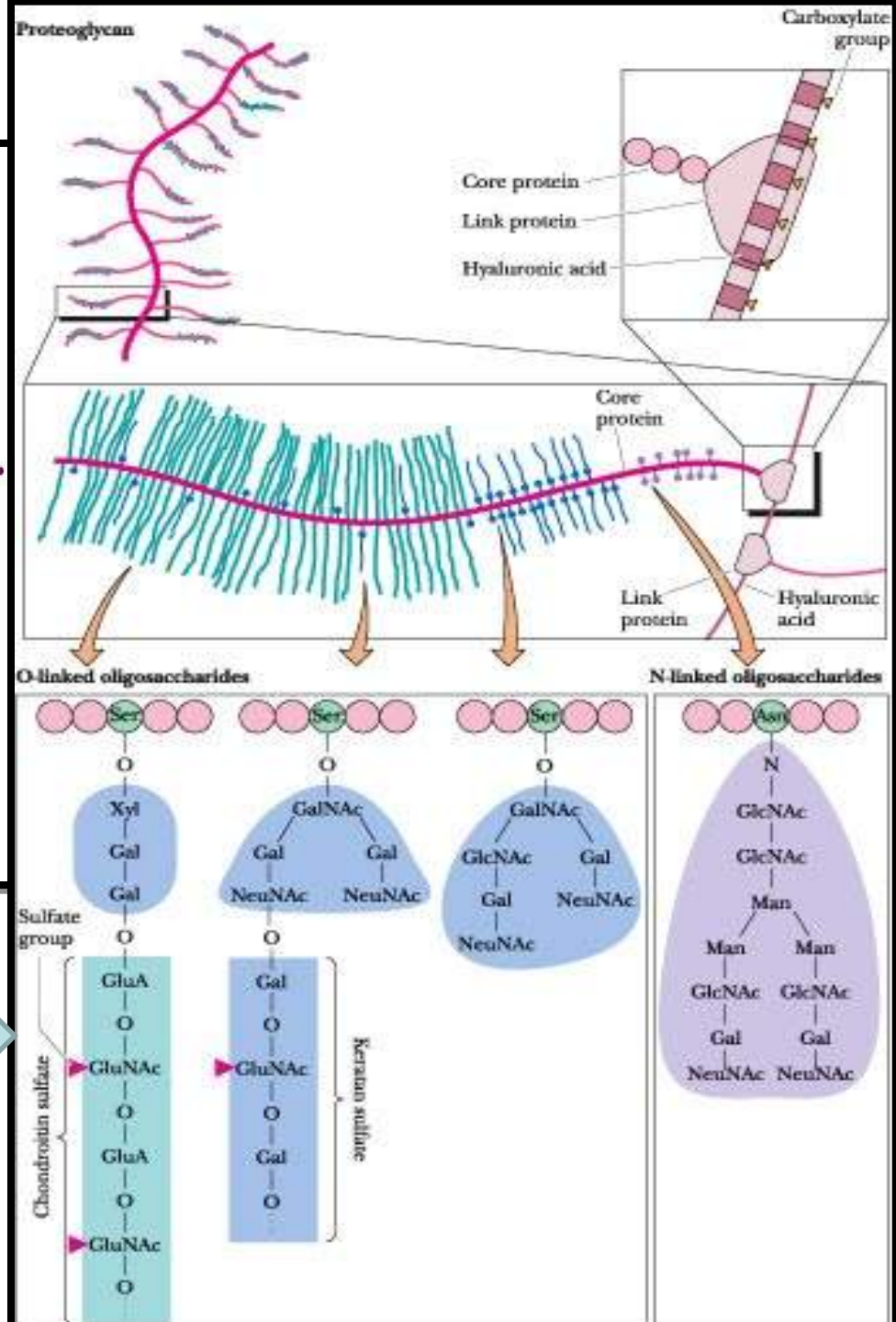


Figure 7-27
Lehninger Principles of Biochemistry, Fifth Edition
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Homopolysaccharide



Heteropolysaccharide



Mucopolysaccharides	Composition	Importance
1. Hyaluronic acid	1. *D*-Glucuronic acid 2. *N*-Acetyl-*D*-glucosamine	Occurs in synovial fluid, skin and vitreous humour.
2. Chondroitin sulphate	1. *D*-Glucouronic acid 2. *N*-Acetyl-*D*-galactosamine sulphate	Components of cartilage, tendons and skin (connective tissue).
3. Dermatan sulphate	1. *L*-Iduronic acid 2. *N*-Acetyl-*D*-galactosamine sulphate	Present in skin
4. Keratan sulphate	1. *D*-galactose 2. *N*-Acetyl *D*-glucosamine sulphate	Components of cartilage and cornea
5. Heparin	1. *L*-Ioduronic acid (sulphated) 2. *N*-Acetyl *D*-glucosamine (sulphated)	Present in most cells especially in liver, lungs and arterial walls.

Chondroitin sulfate

- Chondroitin sulfate is a sulfated glycosaminoglycan (GAG) composed of a chain of alternating sugars (N-acetylgalactosamine and glucuronic acid).
- It is usually found attached to proteins as part of a proteoglycan.
- A chondroitin chain can have over 100 individual sugars, each of which can be sulfated in variable positions and quantities.

Hyaluronic acid

- **Hyaluronic acid** (**HA)** is a non sulfated glycosaminoglycan distributed widely throughout connective, epithelial, and neural tissues.
- It is unique among glycosaminoglycan in that it is non sulfated, forms in the plasma membrane and can be very large, with its molecular weight often reaching the millions.
- One of the chief components of the extracellular matrix, hyaluronan contributes significantly to cell proliferation and migration, and may also be involved in the progression of some malignant tumors.

Keratan sulfate

- Keratan sulfate (KS), also called keratosulfate, is one of several sulfated glycosaminoglycans (structural carbohydrates) that have been found especially in the cornea, cartilage, and bone.
- It is also synthesized in the central nervous system where it participates both in development and in the scar formation following an injury.
- Keratan sulfates are large, highly hydrated molecules which in joints can act as a cushion to absorb mechanical shock.


Thank You